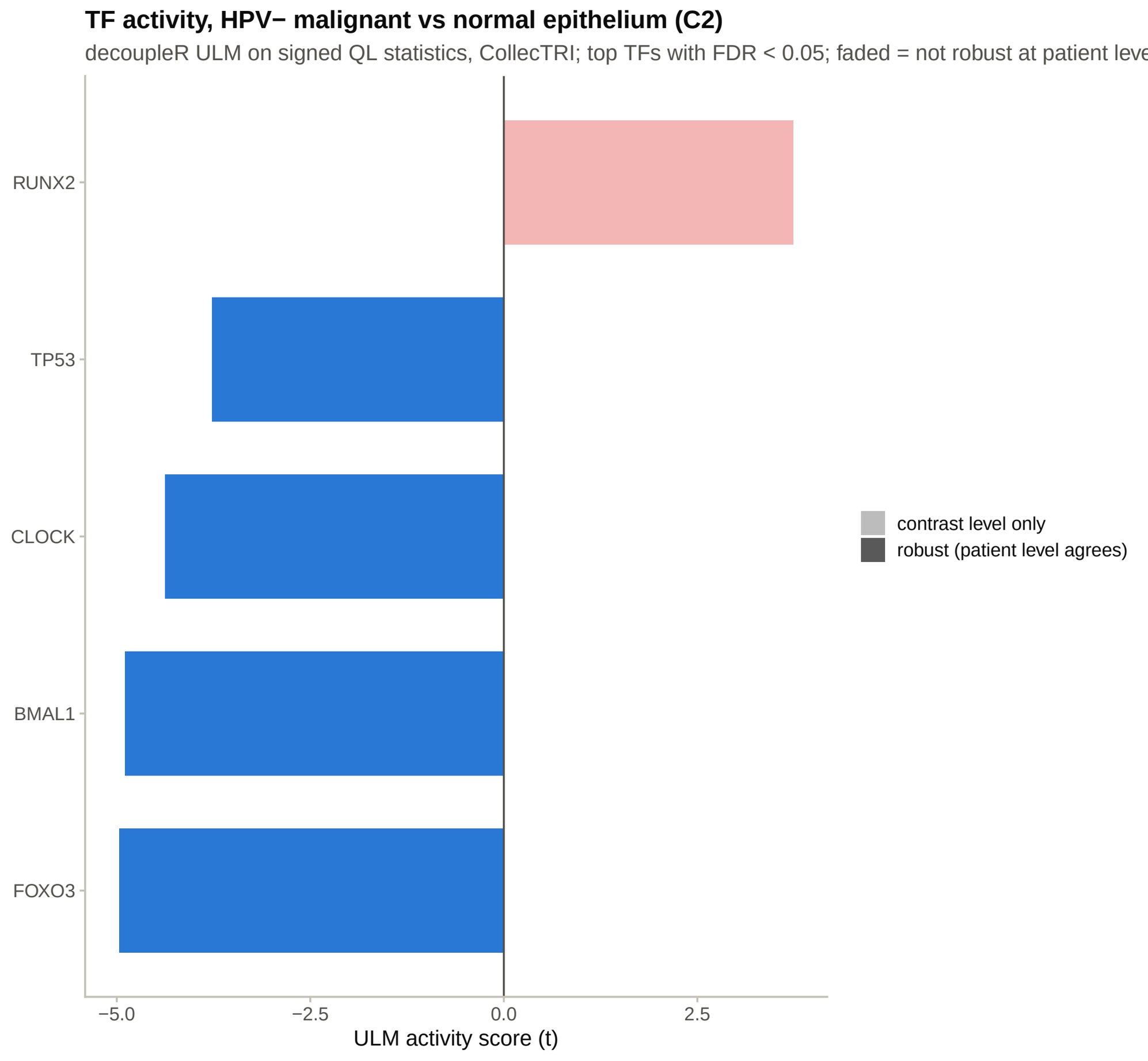
TF activity, HPV− malignant vs normal epithelium (C2)
decoupleR ULM on signed QL statistics, CollecTRI; top TFs with FDR < 0.05; faded = not robust at patient leve
RUNX2
TP53
CLOCK
BMAL1
FOXO3
−5.0
−2.5
0.0
2.5
ULM activity score (t)
contrast level only
robust (patient level agrees)